Buffer or Baseline? Trait Self-Compassion and Employee Well-Being Across Disruption, Telework, and Digital Demands in a National Probability Panel

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This study uses data from the GESIS Panel.pop Standard Edition (ZA5665, Version 62.0.0; GESIS, 2026). Analysis code will be posted in a persistent public repository when the article is published. The study was not preregistered; all hypotheses were specified before substantive models were estimated. This manuscript is posted as a preprint on SSRN. The analyses and interpretations have not been presented or published elsewhere.

During the preparation of this work the author used large language models accessed through Cursor (Anthropic Claude, xAI Grok, OpenAI GPT, and Google Gemini) to draft and revise text, write figure code, and check reported statistics; a word processor with Clippy and autocorrect; and statistical software, so that no coefficient was obtained by hand. The author reviewed, edited, and verified the output and takes full responsibility for the content. These tools are not authors.

Abstract

Conservation of resources theory treats self-compassion (a self-regulatory orientation toward one's own difficulty) as a personal resource that should be associated with employees' level of well-being and attenuate loss when demands arrive. The current study tested that claim three ways in the GESIS Panel.pop, a probability panel of German adults employed in 2018 (N = 3,052): whether prepandemic self-compassion was associated with the course of life satisfaction across the 2020–2021 COVID-19 disruption and with loneliness in 2020; whether it moderated within-person associations of telework share with well-being from 2021 to 2025; and whether it was associated with appraisals of digital technology in 2024. Piecewise growth, random-slope multilevel, and multivariate outcome models were estimated under full-information maximum likelihood, with Bayes factors and repeated nested cross-validation. Life satisfaction fell 6.22 points (0–100 metric) across the disruption at the same rate across levels of self-compassion (BF_{01} = 51.52), and the telework moderation was not supported (BF_{01} = 12.28). The trait was associated with higher life satisfaction, lower 2024 loneliness, and less digital overwhelm. In component-specific models added at the reporting stage, the reverse-scored self-critical items carried these associations (loneliness β = −.30; overwhelm β = −.21), the self-caring items added no unique association, and the estimates survived a six-item neuroticism control (β = −.24 and −.17). Self-compassion measured years earlier was associated with levels of loneliness and digital overwhelm, not with reactivity to an acute disruption, paralleling personality research on life events. Generalization beyond employed adults in one national probability panel requires replication.

Keywords: self-compassion, self-criticism, loneliness, conservation of resources, telework

Generative AI disclosure. During the preparation of this work the author used large language models accessed through Cursor (Anthropic Claude, xAI Grok, OpenAI GPT, and Google Gemini) to draft and revise text, write figure code, and check reported statistics; a word processor with Clippy and autocorrect; and statistical software, so that no coefficient was obtained by hand. The author reviewed, edited, and verified the output and takes full responsibility for the content. These tools are not authors.

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In Germany, 24.2% of employed persons worked from home at least occasionally in 2022, nearly double the 12.8% of 2019 (Statistisches Bundesamt, 2023). Remote, technology-mediated work was imposed on large shares of employees in 2020 (Kniffin et al., 2021). In Germany, employers were legally required to offer home office where feasible from January 2021 to March 2022, with one interruption (Bundesministerium für Arbeit und Soziales, 2021; Gesetz zur Änderung des Infektionsschutzgesetzes und anderer Vorschriften, 2022). The mandates aimed at infection control. Heavier reliance on digital tools was a byproduct, and the technological demands of remote arrangements extend beyond videoconferencing to connectivity expectations and digital monitoring (Booker et al., 2025; Tarafdar et al., 2019). Average changes across this period were small and concealed wide differences between employees. Loneliness rose modestly on average across the disruption, with substantial heterogeneity across samples (Ernst et al., 2022); changes in subjective well-being over the early months of the pandemic showed reliable individual differences (Zacher & Rudolph, 2021); and average associations of telework arrangements with well-being are small (Allen et al., 2015; Gajendran & Harrison, 2007; Gajendran et al., 2024). The open question is which employees maintained life satisfaction and avoided loneliness across the disruption, the elevated telework that followed, and the digital demands employees still reported years later.

The current study examines one candidate resource, self-compassion, and tests one claim about it three ways. Under conservation of resources (COR) theory, a personal resource held before a demand arrives should do two things (Hobfoll, 1989). It should be associated with a person's level of well-being before the demand arrives (a *baseline* difference), and it should

attenuate the loss of well-being when the demand does arrive (a difference in *reactivity*). These propositions are separable, and research on personality and life events shows that the two can diverge. Set points in well-being differ between persons, partly with temperament, and people also differ in whether and how far they adapt to events (Diener et al., 2006). In the British Household Panel Survey, the Big Five did not consistently moderate reaction or adaptation to marriage, childbirth, unemployment, or widowhood (Yap et al., 2012), and that null replicated in an Australian national panel (Anusic et al., 2014). A trait can therefore be a reliable marker of who is better off without being a reliable moderator of who is hurt by an event. The three tests reported here were designed as tests of the resource claim; viewed against this life-events research, they also separate level from reactivity. Hypothesis 1 asks whether prepandemic self-compassion was associated with smaller declines in life satisfaction through the 2020–2021 disruption phase and with lower loneliness in 2020; Hypothesis 2 asks whether it moderated the within-person association of telework share with well-being from late 2021 through 2025; and Hypothesis 3 asks whether it was associated with appraisals of digital technologies in 2024, after prepandemic well-being and technology affinity are controlled.

Self-Compassion and Its Positive and Negative Components

Self-compassion is an orientation toward one's own difficulty comprising self-kindness versus self-judgment, common humanity versus isolation, and mindfulness (i.e., holding painful thoughts and feelings in balanced awareness) versus over-identification (Neff, 2003a). Self-compassion is distinct from related self-constructs in two ways. Unlike self-esteem, it does not require a favorable judgment of the self (Neff, 2003a); unlike self-efficacy (Bandura, 1997), it is not tied to a skill domain. In Neff's (2003a) account, self-compassion concerns how a person responds to difficulty, whatever that difficulty is. It should therefore remain relevant as demands

change (e.g., imposed changes in work location, sustained telework, and unfamiliar digital tools). Meta-analyses have linked higher self-compassion to higher well-being (r = .47 across 79 samples; Zessin et al., 2015) and lower psychopathology (r = −.54; MacBeth & Gumley, 2012). Higher self-compassion has also been associated with a weaker relation of perceived stress to depression and anxiety 6 months later (Stutts et al., 2018). The orientation is trainable in structured programs (Neff & Germer, 2013). In an experimental experience-sampling study of employees, a self-compassionate stance toward oneself as an employee on a given day reduced depletion and raised work self-esteem, with downstream gains in daily engagement and resilience (Jennings et al., 2023). Most of this evidence is cross-sectional or short-lagged, relies on convenience samples, and, in workplace settings, is concentrated in caring professions (Dodson & Heng, 2022). The GESIS Panel's self-compassion module has been analyzed once before, for a 4-month model of stress perception, coping, and affective well-being in the general population (Ewert et al., 2024). To the author's knowledge, the prepandemic assessments have not been used to test whether the resource was associated with employed adults' well-being, loneliness, and technology appraisals years later.

How the scale should be scored is contested, and the dispute bears on what any "self-compassion" effect means. The Self-Compassion Scale (Neff, 2003b) pairs three self-caring facets with three reverse-scored self-critical facets. In a Dutch community sample the two item sets formed separate factors, and López et al. (2015) argued against a total score. Coroiu et al. (2018) reached the same conclusion for the German long form in a general-population sample. A meta-analysis of 18 studies found the negative components more strongly associated with psychopathology than the positive ones and concluded that a reverse-scored total inflates the association (Muris & Petrocchi, 2017). Neff et al. (2019) defended the total on bifactor grounds

across 20 samples. For a prospective resource claim, the scoring question is which item set carries the prediction. If low self-criticism carries the prediction, the "resource" is better described as the absence of a vulnerability, and it is conceptually close to trait neuroticism, the best-documented personality correlate of well-being. The analytic plan scored the short form as a total, following its authors (Raes et al., 2011), and examined its factor structure before use. Component-specific estimates and a neuroticism control were not part of the analytic plan and are labeled exploratory where they appear.

Conservation of Resources

COR theory explains why a resource not tied to any particular demand should be associated with well-being across contexts measured years apart. People strive to retain, protect, and build resources, and stress is a reaction to threatened or actual loss or to a failure of gain after investment (Hobfoll, 1989). Loss is disproportionately salient relative to an equivalent gain (Hobfoll et al., 2018). Two corollaries are relevant to the current study. Those who lack resources are most vulnerable to further loss, and loss spirals develop because they lack the resources to offset that loss (Hobfoll, 1989). Gain spirals are weaker and slower (Hobfoll et al., 2018). Resources also aggregate in caravans, sets of personal resources that are maintained or depleted together (Hobfoll et al., 2018). Self-compassion is proposed as one resource within such a caravan. Specifically, it is the resource that regulates self-judgment when a demand is appraised as personal failure, so that emotional resources remain available for adaptation. In the closest COR treatment, self-compassion was associated with lower exhaustion, directly and through replenished personal resources, in a longitudinal sample of social service providers (Schabram & Heng, 2022). The theory implies that a resource held in advance should preserve well-being when loss is acute, should shape whether a rearranged job is associated with further

loss or with gain, and should remain associated with appraisals of demands years later. The three hypotheses address these implications in turn.

Loneliness and Digital Demands as Criteria

Loneliness and digital overwhelm (i.e., the 2024 item asking whether digital technologies are overwhelming in everyday life) were chosen as criteria because both are work-relevant and both were measured years after the resource. Workplace loneliness has been associated with lower job performance, through reduced approachability and affective commitment, in a time-lagged study of employees and their supervisors (Ozcelik & Barsade, 2018), and with lower job satisfaction and higher burnout across 49 articles (Bryan et al., 2023). Conceptual work treats it as a process arising from organizational antecedents (Wright & Silard, 2021), and reviews place remote work and technology-mediated contact among the work features that can feed it (McCarthy et al., 2026). The GESIS items are the general three-item loneliness scale (Hughes et al., 2004). They measure felt loneliness in general life, which the current study relates to the work arrangement; workplace loneliness itself was not measured. On the technology side, perceived information and communication technology (ICT) demands (e.g., availability, learning, and response expectations) have been associated with strain, stress, and burnout beyond general job demands (Day et al., 2012). The technostress literature treats workplace technology as a demand that can be appraised as threatening or met with coping, including skill development (Tarafdar et al., 2019, 2020). The 2024 items measure such appraisals in everyday life, and workplace technostress itself was not measured.

Overview of the Present Study

The hypothesized predictor is the stable (trait) component of the Self-Compassion Scale–Short Form (SCS-SF) total score, which combines self-caring and reverse-scored self-critical

responding (hereafter trait self-compassion). Its factor structure and stability are examined before the score is used. A two-factor sensitivity analysis then re-estimates each hypothesis test with separate positive and negative composites, and full-sample component-specific models with a neuroticism control, estimated after the hypothesis tests were complete, are reported at the end of the Results. The three tests differ in what they require of the resource: change under an acute demand (Hypothesis 1), moderation of a within-person demand (Hypothesis 2), and association with appraisals of a later demand net of prepandemic well-being and technology affinity (Hypothesis 3). The design timeline and sample flow are shown in Figure 1.

Hypotheses

Under COR, the 2020–2021 window is an acute loss condition in which health threat, social restriction, and an imposed change in how work was done arrived together. Harsh self-judgment would draw on emotional resources that the adjustment itself required, and the disruption is treated as a single acute phase. A resource measured before that window should be associated with the direction of change in life satisfaction during it, beyond earlier well-being and mood. A directional prediction is made only for the disruption phase. Loneliness in 2020, measured on the random half-sample that received the items, tests whether the same trait was associated with felt loneliness during the disruption.

Hypothesis 1. Higher prepandemic self-compassion will be associated with smaller declines in life satisfaction through the 2020–2021 disruption phase and with lower loneliness in 2020, controlling for prepandemic affect.

Most pre-2020 telecommuting evidence is between-person (Allen et al., 2015): it compares employees who usually work from home a great deal with those who usually do not. The within-person contrast compares an employee with the same employee in a year when

telework share departed from that person's own norm. The within-person evidence is limited and equivocal. Employees reported lower stress and work-to-home conflict on telework days than on their own office days in a prepandemic quasi-field experiment (Delanoeije & Verbruggen, 2020), whereas telecommuting predicted later home-to-work interference, without supported indirect paths to distress, in a four-wave pandemic panel (Vander Elst et al., 2026). Under COR, telework intensity is an ambiguous context. A meta-analysis of remote-work intensity specifies opposing indirect paths, gains through autonomy and costs through isolation, with small net benefits on average (Gajendran et al., 2024), and relational costs concentrate at high intensity (Allen et al., 2015; Gajendran & Harrison, 2007). Self-compassion should predict which pattern appears, for life satisfaction, for loneliness in 2024, and for job eudaimonia. Between-person associations of telework level with these criteria are reported as a descriptive validity check. The more precise test is the moderation of within-person change. Hypothesis 2 states the moderation in terms of telework intensity; throughout, intensity is operationalized as telework share, the reported proportion of working time done from home.

Hypothesis 2. Self-compassion will moderate within-person associations of telework intensity with well-being (life satisfaction), loneliness in 2024, and job eudaimonia (learning and personal development), such that higher within-person telework intensity will be associated with preserved or improved well-being, lower loneliness, and higher job eudaimonia among employees higher in self-compassion and with eroded well-being, higher loneliness, and lower job eudaimonia among employees lower in self-compassion.

Finally, if self-compassion is a stable resource, it should still be associated with employees' appraisals of digital technologies once those technologies were ordinary features of daily life and work, years after the 2020 shock. The 2024 demand is repeated unfamiliarity with

tools, which employees may appraise as overwhelming or meet by learning a new technology, and a resource not tied to any particular demand should extend to these appraisals. Technology affinity (i.e., a general liking for technology) is controlled, along with prepandemic well-being, age, and education, so that the association is incremental to liking technology and to earlier well-being. The criteria are overwhelm, the favorability of technology appraisals, the item the panel registry labels "use for job retention," and a single familiarization behavior in the prior 3 months. Hypothesis 3 was framed on that registry label. The item's stem, quoted in the Measures section, measures a national job-loss belief rather than a perceived personal utility, and the hypothesis is retained verbatim with that mismatch disclosed. The behavior is binary, so the association is with the odds of having recently familiarized oneself with a new digital technology.

Hypothesis 3. Trait self-compassion, from the 2018 and 2020 assessments, will be associated with lower digital overwhelm, more favorable technology appraisals (including perceived usefulness for job retention), and greater odds of recent familiarization with a new digital technology in 2024, controlling for prepandemic well-being, age, education, and technology affinity.

Evidence for each hypothesis is reported in graded form rather than as a dichotomous decision: the coefficient with its confidence interval and *p* value, a Bayes factor quantifying relative support for the null and the alternative, and an out-of-sample predictive test.

Method

Transparency and Openness

Data are the GESIS Panel.pop Standard Edition, ZA5665, Version 62.0.0 (GESIS, 2026; https://doi.org/10.4232/1.14772), obtained under a scientific-use agreement that prohibits redistribution of microdata; study materials are the panel's documented questionnaires, available

from the same archive. All variable identifiers are reported so that analyses are reproducible by any researcher with data access. Analysis code (R 4.5.2; lavaan, lme4, glmnet, sandwich, and pROC) will be posted in a persistent public repository when the article is published. The study was not preregistered. Analytic sample size, internal consistency, and rank-order stability of self-compassion were computed before the structural models were estimated. Hypotheses 1–3 and the estimator sequence in this section were specified before any structural coefficient was examined. The Bayes factors, the out-of-sample prediction analyses, the two-factor sensitivity analysis, and the full-sample component-specific models with a neuroticism control were added at the reporting stage. The Results open by naming each of these analyses, and each is marked exploratory in its heading or at first mention. The plan was written for Mplus; the models were estimated in R, and the deviations that this change required are listed at the end of this section.

Assistive Software

The author used large language models accessed through Cursor (Anthropic, 2026; Google, 2026; OpenAI, 2026; xAI, 2026) to draft and revise text, write figure code from cached aggregate files, and check reported statistics; a word processor with Clippy and autocorrect; and statistical software, so that no coefficient was obtained by hand. The author specified the hypotheses, reviewed and verified every number and sentence, and takes responsibility for the content. Prompts and outputs were retained. These tools are not authors.

Participants and Procedure

The GESIS Panel.pop is a probability-based mixed-mode (web and mail) panel of the German-speaking general population, established in 2013 by face-to-face recruitment of adults aged 18–70 from municipal population registers, with bimonthly self-administered surveys thereafter (Bosnjak et al., 2018); its COVID-19 survey program from March 2020 is documented

in a dedicated data paper (Weiß & Stadtmüller, 2023). The panel was refreshed with additional register-recruited cohorts in 2016, 2018, 2021, and 2023, and the current release extends the design beyond the original upper age bound (GESIS, 2026). Because the Self-Compassion Scale was administered in 2018 and 2020, the analytic sample is drawn from the 2013, 2016, and 2018 recruitment cohorts (combined $N = 11{,}797$ persons ever enrolled).

At recruitment and again in each survey wave, panelists are informed of the repeated-survey character and objectives of the study, the voluntary nature of participation, the handling of their survey data, and their data-privacy rights, including the right to deletion. The panel was established before institutional review was required for it and operates under current ethical and data-protection regulations, and demographic and sensitive variables are anonymized or categorized in the scientific-use edition (Weiß & Stadtmüller, 2023). The present study is a secondary analysis of this de-identified scientific-use file and involved no new data collection. [Institutional determination for secondary use of de-identified data to be inserted by the author before submission.]

Of the 11,797 panelists ever enrolled in these cohorts, 5,576 completed the SCS-SF in at least one 2018 wave (4,166 in all three); 5,010 answered the 2018 employment item; and 3,052 reported full- or part-time employment and constitute the primary analytic sample (3,302 under a broader definition adding marginal and irregular employment, reported as a sensitivity analysis). Within the analytic sample, 2,531 completed the 2020 SCS-SF re-measurement, 2,587 provided 2020 life satisfaction, 1,212 provided 2020 loneliness (administered to a random half-sample), 2,089–2,185 provided the individual 2024 loneliness, digitalization, and life-satisfaction items, and 2,137 provided 2025 life satisfaction in the last wave of this release, which represents a retention of 70% from 2018 to 2025.

The mean age of analytic-sample members in 2018 was 47.36 years (*SD* = 11.24; range = 23–75), and 47.7% were women. Educational attainment was high relative to the general population, as expected for an employed subsample: 40.8% held a general or subject-specific university entrance qualification (Abitur), 13.1% an entrance qualification for universities of applied sciences, 33.7% an intermediate school-leaving certificate, and 11.6% a secondary general school-leaving certificate; 60.5% were married and living with their partner.

Attrition across 2018–2025 was addressed with full-information maximum likelihood (FIML) under a missing-at-random assumption and with comparison of completers versus dropouts on focal 2018 variables; a prespecified inverse-probability-of-retention weighting was not obtained (see Deviations From the Planned Estimation). Panelists retained at 2020, 2024, and 2025 differed from those absent by at most one fifth of a standard deviation on 2018 self-compassion, age, gender, education, and 2014 life satisfaction ($|d|$ = 0.00–0.20; Cohen, 1992), with retention modestly selective for older and more educated panelists. Separately from attrition, the GESIS Panel supplies design weights that correct for panelists' unequal probabilities of selection into the panel (wave-specific [wave]za007a; cumulative z000010a–z000010d); their use is described under Analytic Strategy.

Measures

Data set variable names are reported in parentheses throughout this section. Each name begins with a wave prefix whose first letter encodes the field year (b = 2014 through m = 2025); the digitalization module (la) was fielded February–April 2024 and the second loneliness module (lb) May–July 2024.

Self-Compassion (2018, 2020)

The 12-item SCS-SF (Raes et al., 2011) was administered in three 2018 waves (fdcb076a–fdcb087a; fecb135a–fecb146a; ffcb179a–ffcb190a) on a 5-point scale (1 = *does not apply at all*, 5 = *fully applies*). The panel fielded the short form of the German adaptation (SCS-D; Hupfeld & Ruffieux, 2011), with some item wordings adjusted for comprehensibility (Ewert et al., 2024). In 2020, a six-item subset (feelings of inferiority, self-understanding, balanced overview, believing others are happier, mistakes as human nature, giving oneself attention) was administered across four experimental presentation groups (hccx109a–hccx132a; split variable hccx212a), which were pooled item by item. Sample items from the published English short form are "I try to see my failings as part of the human condition" (common humanity) and "When I fail at something important to me I become consumed by feelings of inadequacy" (over-identification, reverse-scored; Raes et al., 2011); the German items as fielded are documented in the panel codebook and the supplement. Negative-subscale items are reverse-scored, and a total composite is formed following Raes et al. (2011), who recommend the short form for total-score use. Because the long-form evidence on that score is divided (Coroiu et al., 2018; López et al., 2015; Neff et al., 2019), the one-factor versus two-factor comparison in Step 1 assessed the scoring question for the present items; the bifactor alternative requires more items per facet than the short form's two-item facets provide. Internal consistency in this sample was ω = .75, .76, and .77 (α = .76, .78, and .77) for the three 2018 administrations and ω = .52 (α = .63) for the six-item 2020 form. Composite rank-order stability was r = .75–.77 within 2018 and r = .63–.64 from 2018 to 2020, consistent with a trait-like resource.

Subjective Well-Being (2014–2025)

Life satisfaction and current happiness were measured with single items administered approximately annually from 2014 through early 2025 (e.g., bazb005a, bazb002a; wave-specific

successors per the variable map); the happiness item was not administered after mid-2020. The life-satisfaction item was administered on an 11-point scale (0 = *extremely dissatisfied* to 10 = *extremely satisfied*) through 2020 and on a 5-point scale (1 = *extremely dissatisfied* to 5 = *extremely satisfied*) from 2021 onward; scores were placed on a common 0–100 percent-of-maximum-possible (POMP) metric (Cohen et al., 1999) for descriptive comparison, and the growth models accommodate the format change explicitly (see Analytic Strategy). Single-item life satisfaction measures show adequate reliability in national panels, including the German Socio-Economic Panel (SOEP; Lucas & Donnellan, 2012), and criterion validity against the Satisfaction With Life Scale comparable to multi-item measurement (Cheung & Lucas, 2014). Prepandemic affect was measured with items in the style of the 12-Item Short Form Health Survey and the Mental Health Inventory (energetic, calm, downhearted; cbaw*–ebaw*) administered bimonthly 2015–2017, with COVID-era administrations (hbcq100a, ibcm243a–244a) used in the disruption phase.

Loneliness (2020, 2024)

Loneliness was measured with three items paralleling the short University of California, Los Angeles (UCLA) Loneliness Scale (Hughes et al., 2004): frequency of lacking companionship, feeling left out, and feeling socially isolated (2020: hfcm217a–hfcm219a, administered to a random half-sample; 2024: lbeo130a–lbeo132a), plus two social-closeness items in 2024 (lbes133a–lbes134a). The 2020 items use a 6-point frequency scale (1 = *never* to 6 = *every day or almost every day*); the 2024 items use a 5-point scale anchored 1 = *very often* to 5 = *never* and are reverse-scored so that higher scores indicate greater loneliness on both occasions. Internal consistency was $\alpha = .83$ (2020) and $\alpha = .79$ (2024). Because anchors differ

across occasions, loneliness was analyzed within occasion (2020 as a Hypothesis 1 outcome; 2024 as a Hypothesis 2 outcome), not as a change score.

Telework Share (2021–2025)

Telework share was measured by the possibility and proportion of working from home, fielded from November 2021 (idcy149a–idcy150a, jacy199a–jacy200a, jbcy222a–jbcy223a, jccy186a–jccy187a) and carried in the annual job module from 2023 (kczg072a–kczg073a, lczg072a–lczg073a, mczg072a–mczg073a; share anchored 1 = *none* to 5 = *more than three quarters of working time*). The COVID-era monthly items in 2020–2021 (hbcy321a, hbcy329a, and successors) asked whether working from home had increased; they do not measure share and are not scored as telework intensity. Share items are routed through employment and home-office filters, yielding 771–989 analytic-sample observations per wave (1,410 persons with at least one report, 6,275 reports in total); respondents without the telework option contribute to the between-person access contrast, and the harmonization of the 2021–2022 formats to the 2023–2025 share metric is reported in the supplement.

Job Characteristics (2014–2025, Annual)

From the recurring job module, learning and personal development (bezg098a → mczg055a) is the job-eudaimonia outcome in Hypothesis 2. The item reads, in the panel's English documentation, "I learn new things at work and have the opportunity for personal development," rated from 1 (*does not apply at all*) to 5 (*totally applies*) with reference to the respondent's current work. The job-eudaimonia label is shorthand for this single item: it indexes the personal-growth facet of eudaimonic functioning as a perceived opportunity, not the broader multidimensional construct.

Digital-Technology Appraisals and Familiarization (2024)

The 2024 digitalization module (fielded February–April 2024) supplies a technology-familiarization behavior in the past 3 months (laem081a), which asks, in the panel's English documentation, "In the last three months, have you familiarized yourself with a digital technology (e.g. technical devices, software programs/apps, etc.) that you have not yet used?" and is scored 1 if the panelist reported familiarizing themselves with a new digital technology in private life, at work, or both, and 0 if not. The module also supplies five evaluations of digital technologies, each rated from 1 (*fully disagree*) to 5 (*totally agree*): making life easier (laem082a), preference for having technology explained by others (laem083a), overwhelm in everyday life (laem084a; "I feel overwhelmed by all the new digital possibilities in my everyday life"), security risk (laem085a), and a belief about the labor-market consequences of digitalization (laem086a). The stem of the last item reads, in translation, "If the opportunities offered by digitalization are not exploited, many jobs will be lost in Germany." The GESIS registry's English variable label for this item is "Use for job retention," and the analytic plan and Hypothesis 3 were framed on that label as perceived usefulness for job retention. The stem measures a national job-loss belief, not a perceived personal utility; the item is therefore reported throughout as the digitalization job-loss belief, Hypothesis 3 is retained verbatim, and the mismatch between the hypothesized construct and the measure is disclosed rather than relabeled. Overwhelm and the job-loss belief are focal single items; the remaining three evaluations form a favorability composite ($\alpha = .29$ in this sample), with scoring detailed in the supplement.

Neuroticism (2014–2018; Added at the Reporting Stage)

A neuroticism control, not part of the analytic plan, was added to test whether the self-critical composite's associations are distinguishable from trait negative affectivity. The primary control is the two-item Big Five Inventory–10 neuroticism scale (nervous; relaxed, reverse-

scored), averaged over each panelist's administrations from 2014 through 2018 (n = 2,754, 90.2% of the analytic sample; two-item $\alpha \approx .47–.54$ per administration). Panelists in the 2018 recruitment cohort have a single contemporaneous 2018 score. Because a two-item control with that reliability removes only part of the construct's variance, the six-item neuroticism scale of the Big Five Inventory–2, administered in 2017 (n = 2,153; panelists recruited in 2018 have no 2017 administration), serves as the stricter sensitivity check. The BFI-10 items are [wave]ze009a (nervous) and [wave]ze004a (relaxed) for the 2014–2018 waves (bd–fd); the BFI-2 items are ecbq045a, ecbq050a, ecbq070a, and, reverse-scored, ecbq055a, ecbq060a, and ecbq065a.

Covariates

Age, gender, and education were taken from the longitudinal demography file; employment status, which defines the analytic sample, comes from the recruitment interview and annual wave modules (e.g., a11d089c, bezh087c, mczh047b). Prepandemic life satisfaction entered the Hypothesis 1 change model as a covariate, and prepandemic well-being entered the Hypothesis 3 outcome regressions as a covariate, so that self-compassion is credited only with variance beyond prepandemic levels.

Analytic Strategy

Latent variable models were estimated in R 4.5.2 (R Core Team, 2025) with lavaan (Rosseel, 2012) under robust maximum likelihood (MLR; Yuan & Bentler, 2000), which combines FIML under a missing-at-random assumption (Enders & Bandalos, 2001) with pseudo-maximum-likelihood incorporation of sampling weights and sandwich standard errors (Asparouhov, 2005); random-slope models were estimated with lme4 (Bates et al., 2015). A single person-level cumulative design weight appropriate to the a1/d1/f1 cohort frame (z000010b) was used, rescaled to a mean of 1; the Kish effective sample size (Kish, 1965) and

the weight's range are reported, and all hypothesis-tested coefficients are reported both weighted and unweighted. For every non-interaction model, the robust χ^2 with scaling factor, the comparative fit index (CFI), the Tucker–Lewis index (TLI), the root-mean-square error of approximation (RMSEA) with 90% CI, and the standardized root-mean-square residual (SRMR) are reported (Hu & Bentler, 1999). Fixed fit-index cutoffs are not treated as pass/fail rules (Marsh et al., 2004); for the self-compassion factor model, dynamic fit-index cutoffs tailored to the fitted model (McNeish & Wolf, 2023) describe misfit. Nested comparisons used the scaled χ^2 difference (Satorra & Bentler, 2010). Each hypothesis-tested path is also accompanied by a Bayes factor computed from the Bayesian information criterion (BIC) comparison of the models with and without that path (Wagenmakers, 2007), so that the relative evidence for the null and the alternative is quantified on a continuous metric rather than only dichotomized at the .05 level. These Bayes factors quantify support for one model relative to the specific alternative compared, not absolute evidence for a hypothesis (Tendeiro et al., 2024). The BIC comparison implies a diffuse unit-information prior on the tested parameter, which favors the simpler model when the effect is small; the factors are therefore benchmarks under that default prior, not evidence under priors chosen for these outcomes (Tendeiro et al., 2024). Reporting follows the APA Journal Article Reporting Standards for quantitative research, including the structural equation modeling module (Appelbaum et al., 2018). The analysis proceeded in four steps, which are described below and mirrored in the Results.

Step 1: Measurement

The 12-item SCS-SF was specified at each of the four occasions, with the six items unadministered in 2020 treated as missing by design (Graham et al., 2006). A one-factor model was compared with a correlated two-factor (positive/negative) model to assess whether the total-

score trait is tenable; McDonald's ω is reported alongside α (McNeish, 2018). Longitudinal invariance was tested configural → metric → scalar across the three 2018 waves, with invariance into 2020 tested on the six common items' loadings and intercepts, and across the 2020 presentation groups before pooling. Evaluation follows Chen (2007): a constraint is judged harmful when $\Delta CFI \geq .010$ is accompanied by $\Delta RMSEA \geq .015$ or the corresponding $\Delta SRMR$ threshold, with partial invariance by freeing the minimal parameter set if required (Byrne et al., 1989). Given at least partial invariance, a trait–state decomposition was estimated (Steyer et al., 2015): a second-order trait factor capturing stable between-person variance across 2018–2020, with occasion-specific state residuals. The trait factor is the focal predictor throughout; its stability is reported because Hypothesis 3's 5- to 7-year lag is defensible only if self-compassion behaves as a trait-like resource. The structural models in Steps 2–4 are diagrammed in Figure S1 of the online supplement.

Step 2: Disruption (Hypothesis 1)

A piecewise latent growth model was estimated for life satisfaction in calendar time (years at wave field midpoints). The slope was allowed to change at two prespecified dates (knots): a predisruption slope through 2019, a disruption slope from the start of 2020 to the start of 2022, and a recovery slope from 2022 through early 2025 (Flora, 2008; Harring et al., 2021). Because the life-satisfaction item moved from an 11-point to a 5-point format between 2020 and 2021, and the disruption window therefore spans the format change, occasions were placed on the POMP metric (Cohen et al., 1999) as the primary harmonization, with two sensitivity analyses: treating post-2020 occasions as ordinal indicators of the same construct (Flora & Curran, 2004), and ending the disruption segment at the final 11-point administration (2020) so that the disruption slope is estimated entirely within one format. The focal parameter is the

regression of the disruption slope on the self-compassion trait, controlling for the intercept, the predisruption slope, prepandemic affect, cohort, and demographic covariates. Occasions before a cohort's recruitment are missing by design, and cohort predicts the intercept and the predisruption slope (Estrada & Ferrer, 2019; Miyazaki & Raudenbush, 2000). Model constraints express the implied total change across the disruption window at ±1 *SD* of the trait, with 95% CIs. Loneliness (2020) is a second outcome regressed on the trait in the same model, with the unsampled half left missing under full-information estimation. Current happiness is measured on a similar schedule; if it is co-administered at enough waves, a two-indicator second-order growth variant is reported (Ferrer et al., 2008), and otherwise happiness is not a focal outcome of this model.

Step 3: Telework (Hypothesis 2)

A random-slope multilevel model was estimated: telework share (2021–2025) was decomposed into between-person and within-person components by person-mean centering (Enders & Tofighi, 2007; see Figure S2 in the online supplement), and the cross-level interaction of the within-person telework slope with the self-compassion trait is the focal coefficient. Share items were modeled only on person-waves where the routing filter applied; persons without a home-office option inform a between-person access contrast rather than being scored as zero share. The intraclass correlation, the within-person *SD* of telework, and the number of persons with at least two on-route reports are reported before any interaction is interpreted (Aguinis et al., 2013). Simple slopes at ±1 *SD* and the Johnson–Neyman region of the trait are reported (Bauer & Curran, 2005; McCabe et al., 2018). Loneliness (2024) and job eudaimonia entered as person-level outcomes of mean telework and of the person-specific slope. The planned latent decomposition with latent moderated structural equations (Klein & Moosbrugger, 2000;

Maslowsky et al., 2015), which avoids the bias that observed person means can introduce into between-person estimates (Preacher et al., 2016), requires Mplus and is deferred to a computational replication (see Deviations From the Planned Estimation).

Step 4: Technology Appraisals (Hypothesis 3)

A multivariate structural model was estimated in which the 2024 technology-appraisal outcomes, with correlated residuals, were regressed on the self-compassion trait, controlling for prepandemic well-being, age, education, and the annual technology-affinity measure ([wave]zf043a–[wave]zf045a; the smartphone-specific item [wave]zf046a was not included because respondents without a smartphone have no valid response) to separate the resource claim from technophilia. The familiarization item (laem081a) was modeled as a binomial logit with heteroskedasticity-consistent standard errors (White, 1980; Zeileis, 2004); an odds ratio (*OR*) with 95% CI is reported. A sensitivity analysis added 2020 well-being to test whether the 2024 association reflects residual well-being rather than the resource.

Missing Data

Four mechanisms were handled and are reported separately (Newman, 2014): panel attrition (FIML with auxiliary correlates of missingness: cohort, age, gender, education, mode, baseline life satisfaction and self-compassion; Graham, 2003); the 2020 loneliness random half-sample (missing completely at random by design; left missing, not imputed); the six SCS-SF items absent in 2020 (missing by design); and off-route telework shares (structurally inapplicable, not imputation targets). A selectivity table compares retained versus absent panelists at 2020, 2024, and 2025 on 2018 focal variables, reporting standardized differences.

Sensitivity Analyses (Prespecified)

Unweighted versus design-weighted estimation; design weight × inverse-probability-of-retention weight (Seaman & White, 2013); 2018-only trait (dropping the 2020 occasion); broad employment definition (N = 3,302); cohorts with at least three pre-2020 life-satisfaction observations; knots moved by one wave; alternative telework harmonizations; pattern-mixture dropout indicator (Enders, 2011); and mode (measurement invariance across web and mail where *n* permits, otherwise a mail-excluded re-estimation).

Sensitivity Analyses (Added at the Reporting Stage)

Seven further checks were added during reporting; the Results name them collectively and mark each exploratory where it is reported. First, a caring-sector stratification: the 2018 industry item (fezg086a) contrasts the education branch and the health and social work branch, the two branches the International Labour Office classifies as care sectors (Addati et al., 2018), with all other branches, because the self-compassion training evidence is concentrated in caring and education professions (Kotera & Van Gordon, 2021); the stratum is industry based, so it includes non-care occupations within those branches and misses care occupations elsewhere. Second, the frequency-weighted telework model was re-estimated with the random telework slope retained, because omitting a random slope makes cross-level interaction tests anti-conservative (Heisig & Schaeffer, 2019). Third, likelihood-ratio and Satterthwaite *p* values were computed beside the normal approximation for the focal cross-level interaction (Luke, 2017). Fourth, the out-of-fold comparison was repeated with unpenalized regression on the same folds, to check that elastic-net shrinkage did not mask incremental signal. Fifth, the 2024 outcome model was refit with a structural-after-measurement estimator, which replaces factor-score regression with a bias-correcting two-step procedure (Rosseel & Loh, 2024). Sixth, because the two-factor analysis indicated that the supported associations were carried by the self-critical

composite, and because the 2024 loneliness estimate had been available only inside the telework model (n = 819), each 2024 outcome was regressed on the two standardized composites simultaneously in the full analytic sample under FIML (estimation N = 3,052), with the Hypothesis 3 covariates, as fully standardized coefficients, unweighted and design-weighted. Seventh, those models were re-estimated with the two-item neuroticism control and, as the stricter check, the six-item 2017 scale. These estimates were obtained after the hypothesis tests and are not tests of a stated prediction.

Out-of-Sample Prediction

The structural models quantify associations in the full sample; they do not test whether the 2018 measures predict the 2024 outcomes for persons the model has not seen (Yarkoni & Westfall, 2017). Each 2024 outcome (digital overwhelm, technology favorability, the digitalization job-loss belief, loneliness, and familiarization) was therefore also submitted to elastic net regression (Zou & Hastie, 2005), implemented in glmnet (Friedman et al., 2010), under repeated nested cross-validation (Pargent et al., 2023): 10 repeats of 5-fold outer cross-validation, with the penalty mixture and strength tuned by 5-fold cross-validation inside each outer training set, so that every observation received an out-of-fold prediction from a model that had no access to it. Out-of-fold predictions were pooled within each repeat to yield one R^2 per repeat (for familiarization, with outcome-stratified folds, one area under the receiver operating characteristic curve, AUC; Hanley & McNeil, 1982; Robin et al., 2011); metrics are means and standard deviations across the 10 repeats. Three feature sets were compared on identical observations per outcome: baseline covariates (age, gender, education, survey mode, cohort, prepandemic affect and life satisfaction, and technology affinity); the covariates plus the 2018 self-compassion composite; and the covariates plus the 12 self-compassion items, each averaged

across the three 2018 administrations. Raw 2018 scores rather than factor scores served as features, so no quantity estimated from the full sample crossed any fold boundary. A single 60/20/20 train/development/test split, run before the cross-validation, is retained as a sensitivity analysis. Cross-validation within one panel quantifies generalization to new persons from this population, not replication in an independent one.

Dynamic structural equation models, cross-lagged specifications that partial out stable between-person variance, fixed-effects regression, and latent profile approaches were considered and not adopted; the rationale for each, in brief, is that Hypothesis 1 concerns who held the resource before the disruption, so between-person resource variance is the quantity of interest rather than a nuisance (full rationale in the supplement).

Deviations From the Planned Estimation

The plan specified Mplus; Mplus was not available, and all models were estimated in R, with five deviations. First, the trait and the structural models were not estimated jointly: the trait enters the growth, multilevel, and 2024 outcome models as a standardized regression factor score from the second-order measurement model, with a 2018-only score, the raw composite, and (in a further sensitivity analysis motivated by the two-factor structure reported below) separate positive and reverse-scored negative composites entered simultaneously as sensitivity predictors; factor-score regression is not equivalent to joint estimation, and its biases depend on the scoring method (Devlieger et al., 2016). Second, the within-person telework decomposition uses observed person-mean centering rather than a latent decomposition, and the cross-level interaction is a fixed product term rather than a latent moderated structural equation. Third, the familiarization logit was estimated separately from the continuous 2024 outcomes because lavaan cannot combine a logit link with full-information continuous estimation. Fourth, design

weights enter lavaan as sampling weights but enter lme4 and the logit model as frequency weights, which does not yield a design-based sandwich estimator; weighted results for those models are labeled sensitivity checks. Fifth, the ordinal treatment of the post-2021 life-satisfaction items, the disruption-ends-in-2020 knot sensitivity, and the pattern-mixture dropout model did not converge or were not estimable and are reported as not obtained rather than interpreted. Robust fit indices for the growth models were numerically unstable, so standard maximum-likelihood indices are reported for them, with MLR standard errors retained. Several smaller prespecified elements were also not obtained: the inverse-probability-of-retention weighting and its product with the design weight, the alternative telework harmonizations, the one-wave knot shifts, and the mode re-estimation. The dynamic fit-index cutoffs were computed during reporting with an archived package build (version and compatibility fix documented in the supplement). ω was computed from standardized loadings, and multilevel *p* values use a normal approximation (Luke, 2017); likelihood-ratio and Satterthwaite approximations for the focal interaction agreed (see Results).

Results

Figure 1 shows the design timeline and sample flow. Table 1 reports descriptive statistics, reliabilities, and correlations among the focal variables. Estimates below are unweighted unless noted. Design-weighted estimates (z000010b, rescaled to a mean of 1, range 0.72–5.09, Kish effective $N = 2{,}894$) are reported as sensitivity checks. The one divergence, involving the frequency-weighted Hypothesis 2 interaction, is noted where it occurs. The following analyses were not part of the analytic plan and were added at the reporting stage: the Bayes factors reported beside each hypothesis test, the out-of-sample prediction, the two-factor and caring-sector sensitivity analyses, and the full-sample component-specific models with a neuroticism

control; each is marked exploratory in its heading or at its first mention, and none is a test of a stated prediction. The seven further checks listed under Sensitivity Analyses in the Method are identified as sensitivity checks where they appear.

Measurement of Self-Compassion

A one-factor model of the 12 items across the three 2018 waves fit poorly, $\chi^2(556, N = 3{,}052) = 6{,}016.90$, $p < .001$, CFI = .848, RMSEA = .057, 90% CI [.056, .058], SRMR = .109. A correlated two-factor model separating the positive and negative item sets provided an appreciable improvement in fit, $\chi^2(539, N = 3{,}052) = 2{,}798.00$, $p < .001$, CFI = .937, RMSEA = .037, 90% CI [.036, .038], SRMR = .072, and the two factors correlated only $r = .32–.39$ within occasion; constraining their correlation to one reproduced the one-factor misfit. Dynamic fit-index cutoffs (McNeish & Wolf, 2023), computed for the single-occasion one-factor model on the first 2018 wave (complete cases, $n = 2{,}516$), placed the most severe misspecification benchmark at CFI = .846, RMSEA = .092, and SRMR = .059; that model's observed values (CFI = .698, RMSEA = .131, SRMR = .131) fell beyond it. The short form is therefore not unidimensional in this sample. The structural models nevertheless use the total-score trait specified in the analytic plan (Table 2); the component-specific estimates reported below address the consequence of that choice.

On the one-factor longitudinal model, metric and scalar constraints were retained across the three 2018 waves under the Chen (2007) joint rule (Table 2), with no intercept freed. Linking the 2020 six-item form, metric invariance was retained (ΔCFI = .006); the scalar constraint reduced CFI by .033 while RMSEA rose by .006 and SRMR by .003, inside the joint rule's bounds but outside a CFI-only threshold of .010, so scalar invariance into 2020 is partial at best and 2020 means are not interpreted. Across the four random 2020 presentation groups, and

across web and mail modes in 2018, metric and scalar constraints were retained ($|\Delta CFI| \leq .007$), which justifies pooling. In the second-order trait–state model, the trait factor accounted for 81%–86% of the variance in the four occasion factors (standardized loadings .90–.93), and occasion-factor correlations were $r = .81–.84$. Standardized regression factor scores from this model are the focal predictor.

Hypothesis 1: Self-Compassion and the Disruption

The piecewise growth model fit the life-satisfaction series, $\chi^2(140, N = 3{,}052) = 424.34$, $p < .001$, CFI = .974, TLI = .966, RMSEA = .026, 90% CI [.023, .029], SRMR = .037, and fit better than a linear alternative (SRMR = .094; Akaike information criterion difference, $\Delta AIC = 465.63$; Akaike, 1974). At average covariates, life satisfaction stood at 70.05 points (0–100 metric), 95% CI [68.35, 71.75], at the start of 2020. The pre-2020 slope was 0.40 points per year, 95% CI [−0.10, 0.90], $p = .119$; the disruption slope was −3.11 points per year, 95% CI [−3.61, −2.61], $p < .001$, an implied loss of 6.22 points across 2020–2021, 95% CI [−7.22, −5.22]; and the recovery slope from 2022 was 0.81 points per year, 95% CI [0.46, 1.15], $p < .001$.

Trait self-compassion was associated with higher life satisfaction at the 2020 knot, $b = 2.16$, 95% CI [1.40, 2.91], $p < .001$, $\beta = .14$, and with a more positive pre-2020 slope, $b = 0.25$, 95% CI [0.05, 0.45], $p = .015$, $\beta = .12$, but it was not significantly associated with the disruption slope, $b = −0.10$, 95% CI [−0.71, 0.51], $p = .746$, $\beta = −.02$, or with the recovery slope, $b = −0.29$, 95% CI [−0.69, 0.11], $p = .154$, $\beta = −.09$. The implied disruption-window change was −6.02 points at −1 *SD* of the trait, 95% CI [−7.50, −4.54], and −6.42 points at +1 *SD*, 95% CI [−8.09, −4.76] (Figure 2; Table 3). The trait was also not significantly associated with 2020 loneliness, $b = −0.06$, 95% CI [−0.16, 0.03], $p = .166$, $\beta = −.05$ ($n = 1{,}212$). Exploratory Bayes factors favored the models without these paths, $BF_{01} = 51.52$ for the disruption slope and $BF_{01} = 18.42$ for 2020

loneliness. These results do not support Hypothesis 1. The nonsignificant disruption-slope coefficient was unchanged with the 2018-only trait ($b = -0.09$, 95% CI [−0.70, 0.51], $p = .763$), the raw 2018 composite ($b = -0.20$, 95% CI [−0.83, 0.44], $p = .545$), design weights ($b = -0.05$, 95% CI [−0.68, 0.57], $p = .866$), the 2018 recruitment cohort excluded ($b = -0.20$, 95% CI [−0.87, 0.47], $p = .551$), and the broad employment definition ($b = -0.20$, 95% CI [−0.80, 0.39], $p = .507$; $N = 3{,}302$).

Hypothesis 2: Self-Compassion and Telework

Among the 1,410 participants with at least one on-route telework report (6,275 reports; 1,164 with two or more), the intraclass correlation of telework share was .75, and the within-person *SD* of person-mean-centered share was 0.69 (person-year counts in Table 4). In the unmoderated random-slope model, a within-person deviation in telework share was not associated with life satisfaction, $b = -0.85$ points per share point, 95% CI [−1.91, 0.21], $p = .118$, and the person-mean share was likewise not associated, $b = 0.52$, 95% CI [−0.17, 1.21], $p = .140$; trait self-compassion was associated with higher life satisfaction in this model, $b = 1.58$, 95% CI [0.49, 2.68], $p = .005$. Having any recorded home-office access, versus none, was associated with higher mean life satisfaction, $b = 1.61$, 95% CI [0.18, 3.04], $p = .027$ ($n = 1{,}828$).

The cross-level interaction of trait self-compassion with within-person telework was not significant, $b = -0.96$, 95% CI [−2.01, 0.10], $p = .076$ (Table 4; the likelihood-ratio and Satterthwaite approximations gave $p = .076$ and $p = .077$), which does not support Hypothesis 2. The simple slopes ran opposite to the hypothesized pattern: the within-person telework slope was near zero at −1 *SD* of the trait ($b = 0.10$, 95% CI [−1.39, 1.59], $p = .893$) and negative at +1 *SD* ($b = -1.81$, 95% CI [−3.31, −0.31], $p = .018$), with the Johnson–Neyman band excluding zero above 0.25 *SD* of the trait (Figure 3); because the interaction was not significant, the pattern is

descriptive. The exploratory Bayes factor favored the model without the interaction, BF_{01} = 12.28. A frequency-weighted sensitivity model did reach significance (b = −1.02, 95% CI [−1.98, −0.06], p = .037), but it is not a design-based estimator and omitted the random telework slope, which makes cross-level interaction tests anti-conservative (Heisig & Schaeffer, 2019); with the random slope retained, the weighted interaction was b = −1.03, 95% CI [−2.09, 0.04], p = .059, so the primary random-slope estimate is the basis for evaluating Hypothesis 2. Neither person-mean telework nor the person-specific telework slope predicted 2024 loneliness (n = 819; b = 0.01, 95% CI [−0.02, 0.04], p = .627, and b = 0.08, 95% CI [−0.12, 0.29], p = .430) or job eudaimonia (n = 735; b = 0.02, 95% CI [−0.03, 0.07], p = .380, and b = −0.12, 95% CI [−0.45, 0.20], p = .460). The planned trait moderations of these person-level paths were estimated and did not differ from zero (loneliness: person-mean × trait b = −0.02, p = .248, and slope × trait b = 0.00, p = .971; job eudaimonia: b = 0.01, p = .798, and b = 0.14, p = .409; full estimates with confidence intervals in the online supplement), so the person-level coefficients above are simple effects at the mean of the standardized trait. Trait self-compassion was unrelated to job eudaimonia, b = −0.01, 95% CI [−0.08, 0.07], p = .854; its association with 2024 loneliness, the largest in the study, is reported in the next section.

Self-Compassion and Loneliness in 2024

The strongest association in the study was not a hypothesized path. Trait self-compassion was associated with lower loneliness in 2024, b = −0.19, 95% CI [−0.24, −0.14], p < .001 (n = 819), estimated as the person-level association in the Hypothesis 2 model; the hypothesis for this outcome concerned the telework moderation reported above, not this direct path, so the association is reported as estimated rather than as a test of a stated prediction. The zero-order correlation between the 2018 composite and 2024 loneliness was r = −.36 (Table 1), larger than

for any other outcome, and it is the only outcome for which self-compassion carried a substantial out-of-fold increment, $\Delta R^2 = .036$ (see Out-of-Sample Prediction). In the two-factor sensitivity analysis it was carried by the reverse-scored self-critical items, $b = -0.18$, 95% CI [−0.23, −0.13], $p < .001$, and not by the self-caring items, $b = -0.01$, 95% CI [−0.06, 0.03], $p = .615$. The same trait was unrelated to loneliness during the 2020 disruption but was associated with loneliness 4 years later, although the response anchors changed between the two administrations, so the magnitudes are not directly comparable. Because the $n = 819$ estimate is conditioned on having telework reports, an exploratory full-sample specification, estimated after the hypothesis tests, is reported below.

Hypothesis 3: Self-Compassion and 2024 Technology Appraisals

Controlling for prepandemic well-being, age, education, and technology affinity, trait self-compassion was associated with lower digital overwhelm, $b = -0.20$, 95% CI [−0.24, −0.16], $p < .001$, $\beta = -.20$, and with more favorable technology appraisals, $b = 0.06$, 95% CI [0.04, 0.09], $p < .001$, $\beta = .10$. It was not significantly associated with the digitalization job-loss belief (the item hypothesized, on its registry label, as perceived usefulness for job retention; see Measures), $b = 0.02$, 95% CI [−0.03, 0.07], $p = .405$, or with the odds of recent familiarization with a new digital technology, $OR = 0.98$, 95% CI [0.88, 1.08], $p = .641$ ($n = 2{,}066$; 978 events). Exploratory Bayes factors favored the alternative for overwhelm and favorability, $BF_{10} = 1.44 \times 10^{19}$ and 3,179.03, and favored the null for the job-loss belief and familiarization, $BF_{01} = 38.01$ and 40.73. These results provide partial support for Hypothesis 3, with two of four outcomes supported (Table 5; Figure 4). The pattern of results was unchanged when 2020 life satisfaction was added, the 2018-only trait was substituted, or design weights were applied. A structural-after-measurement refit, which replaces factor-score regression with a bias-correcting two-step

estimator (Rosseel & Loh, 2024), gave nearly identical standardized coefficients (overwhelm β = −.22, favorability β = .11, job-loss belief β = .02).

Sensitivity Analyses and Models Not Obtained

Three prespecified sensitivity analyses could not be estimated: the ordinal treatment of the post-2021 life-satisfaction items (rank deficient), and the disruption-segment-ends-in-2020 knot variant and the pattern-mixture dropout model (nonconvergence); the smaller elements listed under Deviations From the Planned Estimation were likewise not obtained. A model fixing residual variances from adjacent-wave reliabilities (Hayduk & Littvay, 2012) left the disruption slope essentially unchanged (b = −0.09, 95% CI [−0.70, 0.52], p = .777), and current happiness could not serve as a second well-being indicator because the item was discontinued in mid-2020.

Out-of-Sample Prediction (Exploratory)

Across 10 repeats of 5-fold nested cross-validation (1,528–1,596 complete cases per outcome), baseline covariates alone predicted the 2024 outcomes at out-of-fold R^2 = .099 (job-loss belief) to .268 (favorability), with familiarization AUC = .682; across-repeat standard deviations were at most 0.004. Adding the 2018 self-compassion composite improved out-of-fold prediction for 2024 loneliness, ΔR^2 = .036 (SD = 0.002), and for digital overwhelm, ΔR^2 = .008 (SD = 0.001); it did not improve technology favorability, the job-loss belief, or familiarization (increments ≤ .001 in absolute value). The 12 item-level scores produced the largest increments for loneliness, ΔR^2 = .056 (SD = 0.004), and overwhelm, ΔR^2 = .021 (SD = 0.003), with increments of .006 or less elsewhere. Unpenalized regressions on the same folds gave essentially the same increments, so the near-zero increments are not an artifact of elastic-net shrinkage; the single train/development/test split run before the cross-validation agreed for loneliness and disagreed elsewhere, which illustrates the split-to-split noise the repeated design

averages out. The repeat standard deviations describe stability across fold assignments, not sampling uncertainty, and the complete-case prediction sample is narrower than the structural models' analytic sample. Therefore self-compassion carried predictive signal beyond the covariates for loneliness and, more modestly, for digital overwhelm, whereas favorability, the job-loss belief, and familiarization were predicted as well by the baseline covariates alone (per-outcome and per-repeat metrics in the supplement).

Two-Factor Sensitivity Analysis (Exploratory)

Because the positive and negative item sets were only weakly correlated, each hypothesis test was re-estimated with two standardized composites entered simultaneously in place of the total-score trait: a positive composite (self-caring items; cross-wave α = .70) and a reverse-scored negative composite (self-critical items; α = .90), which correlated $r = .26$. The Hypothesis 1 nulls held for both composites: neither predicted the disruption slope (positive, $b = -0.36$, 95% CI [−0.87, 0.15], $p = .170$; reversed negative, $b = 0.08$, 95% CI [−0.50, 0.66], $p = .783$) or 2020 loneliness ($p = .236$ and $p = .128$). In the Hypothesis 2 life-satisfaction model, the within-person telework interaction was statistically significant for the reversed negative composite, $b = -1.24$, 95% CI [−2.39, −0.09], $p = .034$, and not for the positive composite, $b = 0.48$, 95% CI [−0.66, 1.63], $p = .411$; the significant interaction matches the direction of the weighted total-score interaction reported above, opposite to the hypothesized buffering. The Hypothesis 3 associations divided between the two composites. The reversed negative composite carried the overwhelm association, β = −.21, 95% CI [−.26, −.17], $p < .001$. The positive composite was the only composite associated with the digitalization job-loss belief, β = .06, 95% CI [.01, .10], $p = .010$, an association the total score did not detect and one that is qualified below. Component-specific estimates for the favorability composite are not interpreted: the three-item composite had

α = .29, the two composites' zero-order correlations with it were near zero (−.02 and .07), and their simultaneous estimates diverged from those zero-orders in the pattern of mutual suppression, which on a composite of that reliability is not a substantive difference between the item sets. Familiarization odds were unrelated to either composite (p = .112 and p = .240). Interactions for the secondary Hypothesis 2 outcomes showed no pattern consistent across the two composites; full estimates appear in the online supplement. Therefore the supported associations were carried primarily by low self-critical responding rather than by self-caring responding.

Component-Specific and Neuroticism-Adjusted Estimates (Added at the Reporting Stage)

The estimates in this section were obtained after the hypothesis tests, in response to the two-factor pattern above, and are not tests of a stated prediction. The hypothesized trait factor score correlated r = .98 with the reverse-scored self-critical composite and r = .38 with the self-caring composite, so the total-score associations reported for Hypotheses 1–3 were, in effect, associations of low self-criticism. Coefficients below are fully standardized, from models estimated on the full analytic sample under FIML (estimation N = 3,052) with the Hypothesis 3 covariates, and are not on the metric of the unstandardized coefficients above.

For 2024 loneliness (n = 2,174 observed), the reverse-scored self-critical composite (higher scores = less self-critical responding) was associated with lower loneliness, β = −.297, p < .001 (zero-order r = −.398), and the self-caring composite entered simultaneously was not, β = .010; the self-caring composite's zero-order association, r = −.113, was shared with the self-critical composite. The self-critical estimate was β = −.275 with the two-item neuroticism control, β = −.239 with the six-item 2017 scale (n = 1,616), and β = −.295 design-weighted. For digital overwhelm (n = 2,183), the self-critical composite was associated with less overwhelm, β

= −.213, $p < .001$; β = −.177 with the two-item control, β = −.169 with the six-item scale, and β = −.207 design-weighted. (Three decimals are given here so that the exact estimates can be matched to the analysis output; rounded values are used elsewhere.) The neuroticism control reduced both associations by at most one fifth and removed neither; because the two-item control is weak (α ≈ .47–.54), the six-item check is the better estimate of what neuroticism accounts for. For the digitalization job-loss belief, the self-caring composite showed a small positive association, β = .06, $p = .010$, the same whether entered alone or with the self-critical composite, that fell to β = .05, $p = .071$, under the six-item control; given its size, its instability under the stricter control, and the number of tests in this family, it is reported as a small association that did not hold under the stricter control, not as a finding. Familiarization odds were unrelated to either composite in every adjusted model.

Caring-Sector Sensitivity Analysis (Exploratory)

Among the 2,843 participants with a valid 2018 industry branch (704 in the care sectors, 2,139 elsewhere), trait self-compassion did not differ between strata ($d = 0.03$), and the trait × sector interaction was near zero for digital overwhelm, $b = 0.00$, 95% CI [−0.09, 0.09], $p = .995$, and for 2024 loneliness, $b = 0.00$, 95% CI [−0.11, 0.12], $p = .976$, with the same within-stratum slope ($b = −0.20$, $p < .001$) in both strata for both outcomes. Only the favorability interaction was significant, $b = −0.07$, $p = .013$, with the association present outside the care sectors and absent within them (full estimates in the supplement). The study's two strongest associations were therefore the same size inside and outside the settings in which most self-compassion research has been conducted.

Discussion

Findings of the current study separated level from reactivity across three tests of one resource claim. Self-compassion measured in 2018 did not predict the course of life satisfaction across the 2020–2021 disruption, loneliness during it, or the within-person association of telework share with life satisfaction from late 2021 through 2025. The Bayes factors favored the models without the disruption and 2020 loneliness paths (BF_{01} = 51.52 and 18.42); the factor for the telework interaction was weaker (BF_{01} = 12.28) and is examined below. Hypotheses 1 and 2 were therefore not supported. The same trait was associated with higher life satisfaction at the onset of the disruption, with less digital overwhelm and more favorable technology appraisals in 2024, and, in a path that was not hypothesized, with lower loneliness in 2024, the largest association in the study. Hypothesis 3 was partially supported. In the exploratory component-specific models, the reverse-scored self-critical items carried the loneliness and overwhelm associations in the full sample (β = −.30 and −.21), the self-caring items added no unique association, and the estimates were reduced but not removed by a neuroticism control (β = −.24 and −.17 under the six-item scale).

The dissociation is between levels and change, a pattern with precedent in the life-events literature reviewed in the introduction, in which broad traits predicted a person's level of well-being without consistently moderating reaction or adaptation to events (Anusic et al., 2014; Diener et al., 2006; Yap et al., 2012). The current results extend that pattern from broad traits and discrete personal events to a narrower self-regulatory resource and a population-wide disruption. Prospective work on the early pandemic found reliable individual differences in well-being change (Zacher & Rudolph, 2021). Self-compassion measured 2 years earlier was not associated with those differences in this sample. A resource-based examination of the same period in a U.S. probability panel likewise found no buffering. Adults with more education, whom a resource

account would expect to be better buffered, showed larger declines in life satisfaction and larger rises in depressive symptoms, and perceived resources did not explain that difference (Wanberg et al., 2020).

In COR terms, the caravan proposition (Hobfoll et al., 2018) is compatible with this pattern, but the design measured no second resource and cannot test it. The theory's stronger prediction, that held resources attenuate loss under acute demand (Hobfoll, 1989), was not supported: the decline in life satisfaction was broad and nearly uniform across levels of self-compassion. The first boundary condition concerns the kind of demand. A population-wide, externally imposed disruption may be a weak test of a resource whose proposed mechanism is the appraisal of personal failure, because the 2020 decline may have carried little self-evaluative content to regulate. A three-wave occupational study is consistent with this boundary: self-compassion weakened the prospective association of felt work stressors with depressive and anxiety symptoms, and its authors contrast that moderation with null results for objectively indexed stressors (de Krijger et al., 2025). The 2024 results fit this account: feeling overwhelmed by digital technology is a self-referent appraisal, and the familiarization behavior, which is an action, showed no association.

The telework test suggests a second boundary condition. Telework intensity is an ambiguous demand, with opposing paths through autonomy and isolation (Gajendran et al., 2024). The within-person moderation was not significant, and its direction ran opposite to the hypothesized buffering: higher within-person telework was associated with lower life satisfaction among employees higher in self-compassion. That pattern reached $p = .034$ for the self-critical composite in the two-factor sensitivity analysis and should still be read as descriptive. The Bayes factor against the interaction rests on the diffuse BIC prior and a penalty counted in person-years.

The likelihood-ratio statistic was $\chi^2(1) = 3.14$, and a person-level penalty yields a factor near 7, so the evidence against the interaction is weaker than for the near-zero paths. The closest within-person study found buffering, with self-compassion weakening the daily association of work loneliness with strain during the pandemic's first year (Andel et al., 2021), so whether self-compassion moderates anything within persons remains open.

The measurement result bears on how the supported associations should be interpreted. The positive and negative item sets correlated only $r = .32–.39$, as the structural critiques of the scale predict (Coroiu et al., 2018; López et al., 2015); the trait factor correlated $r = .98$ with the self-critical composite; and the associations the total score detected were the self-critical composite's. This is the pattern Muris and Petrocchi (2017) found across 18 studies for psychopathology, here in a prospective panel for loneliness and a technology appraisal. Therefore, prospective "self-compassion" associations estimated from total scores, including those for Hypothesis 3 in this paper, may be low self-criticism associations, and the element that does the theoretical work in a COR account is the absence of harsh self-judgment. The chief rival explanation is that low self-criticism is low neuroticism under another name. The two-item control and the six-item 2017 scale reduced the loneliness and overwhelm coefficients without removing them (see Results), so the self-critical composite carried association beyond neuroticism as measured. A two-item control cannot settle the question, and the six-item check is the better estimate of residual confounding. Future tests of self-compassion as a resource should score the positive and negative components separately, state which one the theory needs, and include a full neuroticism measure (Muris & Petrocchi, 2017); for the contrary scoring view, see Neff et al. (2019).

Practical Implications

The design is correlational: self-compassion was not manipulated, and the results do not indicate what changing it would do. Within that scope, two implications follow. The first is descriptive. A brief measure administered 6 years earlier was associated with loneliness and digital overwhelm in 2024, with employees lower in prepandemic self-compassion reporting more of both, net of technology affinity and earlier well-being in the primary models; in exploratory component models the self-critical associations were reduced but not removed after a neuroticism control; in out-of-fold predictions the composite added signal beyond the prespecified baseline covariates for exactly those two outcomes. Both outcomes matter at work: loneliness is associated with lower performance, lower satisfaction, and burnout (Bryan et al., 2023; Ozcelik & Barsade, 2018), and perceived ICT demands with strain beyond general job demands (Day et al., 2012). The associations held equally inside and outside the care sectors in which most self-compassion research has been conducted. The second implication concerns intervention, and the expectations should be modest. Self-compassion can be raised by structured programs (Neff & Germer, 2013), and training has shown effects on work-related well-being mostly in pre–post designs in caring professions (Kotera & Van Gordon, 2021). At the daily level, the same stance was followed by lower depletion and higher work self-esteem in the experimental experience-sampling study cited earlier (Jennings et al., 2023). What the current results add is a target: if the prospective association runs through the self-critical items, programs aimed at reducing harsh self-judgment are the ones the panel evidence supports. The organizational antecedents of loneliness also warrant attention (Wright & Silard, 2021), and whether such programs would change later loneliness or digital overwhelm has not been tested.

Limitations

There are several limitations in the current study. First, the design is observational. Self-compassion was measured before the outcomes, and the controls for prepandemic well-being and neuroticism address some rival explanations. However, the primary neuroticism control is a two-item scale ($\alpha \approx .47–.54$), the six-item check is available for about half the sample, and unmeasured third variables (prior adversity, health) could produce the same pattern. Every claim is associational, and the language of resource preservation refers to the hypothesized mechanism, not an established one.

Second, the sample's composition constrains generalization. Roberts et al. (2020) recommend that authors justify a sample's demographic composition as they would its size. This sample is a register-recruited probability sample of German-speaking adults, aged 18–70 at recruitment, who were employed in 2018. It is more educated than the general population, as an employed subsample would be, and it was observed across one disruption and one country's telework mandates. It was chosen because it is the only probability panel known to the author with self-compassion measured before 2020, not because its composition represents employees elsewhere. Migration background and ethnicity were not modeled, and the sequence of an abrupt involuntary disruption followed by prolonged elevated telework may not generalize to other labor markets or to workplace changes that did not coincide with a pandemic. The recovery segment describes life satisfaction in this panel, not German mental health broadly, where repeated cross-sectional screening found depressive-symptom elevations persisting into 2022 (Mauz et al., 2023).

Third, measurement constrains sensitivity and comparability. All measures were self-report, and well-being rests on single items whose reliability limits sensitivity (Lucas & Donnellan, 2012). The life-satisfaction format changed from 11 to 5 points in 2021, inside the

disruption window, and the planned ordinal sensitivity could not be estimated, so the disruption slope rests on the POMP harmonization alone. The loneliness anchors changed between 2020 and 2024, so the two loneliness magnitudes are not comparable, and 2020 loneliness was administered to a random half-sample. The short form was not unidimensional, the 2020 occasion is a six-item form with $\omega = .52$, the favorability composite had $\alpha = .29$, and the digitalization job-loss item measures a national belief rather than the perceived utility the hypothesis named. Telework share was not measured in 2020, so the moderation window begins in late 2021. The panel contains no items on artificial intelligence or workplace technostress. The estimation deviations listed in the Method (factor-score regression instead of joint estimation, observed instead of latent within-person decomposition) can bias structural coefficients, in directions that depend on the scoring method and the location of the effect in the model (Devlieger et al., 2016; Preacher et al., 2016). The null moderation is therefore conditional on these approximations, and the latent versions remain to be estimated.

Fourth, attrition remains a limitation. Although 70% of the analytic sample provided data in 2025, the missing-data handling assumes that data are missing at random conditional on modeled variables, and panelists recruited in 2021 and 2023, who have no self-compassion measure, are not included.

Future Directions

The component-specific result is the one most in need of independent replication. Probability panels with prepandemic personality and later loneliness items could test, as a registered report, whether the self-critical composite predicts loneliness and technology appraisals with a full neuroticism measure in the model. An experimental or intervention design is needed to test whether reducing self-criticism changes the appraisal outcomes the panel

supported (Neff & Germer, 2013), and a randomized trial with a workplace-change stressor would also test the buffering the measured trait did not show. Probability panels could also field artificial intelligence–specific exposure and appraisal items. Finally, denser measurement around a future disruption would show whether state fluctuations in self-compassion, beyond the stable trait, track resource loss and recovery within persons.

Conclusion

In a decade of probability-panel data on German employees, self-compassion measured in 2018 was associated with levels of life satisfaction, with lower loneliness in 2024, and with less digital overwhelm 6 years later. It was not associated with the rate of loss during the COVID-19 disruption or with the within-person consequences of telework. The associations that held were carried by the reverse-scored self-critical items and survived a neuroticism control. Self-compassion, as the short form measures it, indicated which employees reported greater well-being years later; it did not indicate who was protected when the demand arrived. Tests of self-compassion as a resource should state which component they mean; in the current panel, the supported associations were concentrated in the self-critical composite.

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Table 1

Descriptive Statistics and Intercorrelations Among Focal Variables

Variable	*n*	*M*	*SD*	1	2	3	4	5	6	7	8	9	10	11	12	13
1. Self-compassion 2018	3,052	3.31	0.47													
2. Self-compassion 2020	2,531	3.38	0.55	.70**												
3. Life satisfaction 2014	1,611	69.99	19.00	.24**	.18**											
4. Life satisfaction 2020	2,587	71.21	18.99	.32**	.33**	.41**										
5. Life satisfaction 2024	2,173	66.97	23.23	.19**	.19**	.23**	.36**									
6. Life satisfaction 2025	2,137	67.88	23.16	.19**	.17**	.25**	.37**	.42**								
7. Loneliness 2020	1,212	2.60	1.30	−.15**	−.14**	−.13**	−.18**	−.02	−.11**							
8. Loneliness 2024	2,174	2.04	0.70	−.36**	−.31**	−.22**	−.32**	−.28**	−.25**	.32**						
9. Telework share	1,410	2.62	1.43	.06*	.07**	−.01	.00	.02	.01	.05	−.01					
10. Digital overwhelm 2024	2,183	2.48	0.98	−.14**	−.15**	−.06	−.08**	−.11**	−.10**	.09**	.14**	−.11**				
11. Digitalization job-loss belief 2024	2,170	3.42	1.02	.08**	.06**	.07*	.11**	.10**	.07**	.00	−.04	.10**	−.17**			
12. Familiarization 2024	2,089	.47	0.50	.00	.00	.03	.03	.04	.04	.01	−.02	.08**	−.18**	.12**		
13. Age 2018	3,052	47.36	11.24	.18**	.10**	.02	.04*	.02	.02	−.12**	−.12**	−.03	.17**	.02	−.13**	
14. Gender	3,052	1.48	0.50	−.03	.01	−.03	.02	.01	.00	.07*	.03	−.08**	.16**	−.17**	−.11**	−.02

Note. Correlations are below the diagonal; the diagonal is blank. Life satisfaction is percent of maximum possible (POMP; 0–100). Self-compassion 2018 is the mean of a panelist's available 2018 occasions (up to three); self-compassion 2020 is the six-item score. Loneliness 2020 is on a 1–6 frequency scale, and loneliness 2024 is on a reversed 1–5 scale. Telework share is the person mean on a 1–5 share scale. The digitalization job-loss belief is the item the GESIS registry labels "Use for job retention" (laem086a); see Measures for its stem. Familiarization is binary (1 = familiarized with a new digital technology in the past 3 months in any context, 0 = none). Gender is coded 1 = male and 2 = female. Age was recorded in 2018. Pairwise *n*s ranged from 654 to 3,052. Two-tailed

probabilities used $t = r\sqrt{((n-2)/(1-r^2))}$ with $n-2$ degrees of freedom. Correlations are rounded to two decimals, so two entries displayed as .05 need not share a significance star. *$p < .05$. **$p < .01$.

Table 2

Measurement Models and Longitudinal Invariance of the Self-Compassion Scale–Short Form

Model	χ^2	*df*	CFI	RMSEA	SRMR	ΔCFI	ΔRMSEA
Factor structure, 2018							
One-factor	6,016.90	556	.848	.057	.109		
Two-factor	2,798.00	539	.937	.037	.072	.089	−.020
Two-factor, interfactor *r* = 1	5,915.40	545	.859	.064	.112	−.078	.027
Invariance across the 2018 occasions							
Configural (one-factor model)	6,016.90	556	.848	.057	.109		
Metric	6,082.30	576	.856	.063	.110	.008	.006
Scalar	6,343.10	600	.851	.062	.110	−.005	−.001
Linking the 2020 occasion							
Configural	7,646.90	760	.860	.054	.105		
Metric	6,693.40	784	.866	.055	.105	.006	.001
Scalar	8,113.70	814	.833	.061	.108	−.033	.006
2020 presentation groups							
Configural	515.20	36	.766	.134	.067		
Metric	399.10	51	.763	.113	.068	−.003	−.021
Scalar	425.20	66	.764	.099	.070	.001	−.014
Survey mode							
Configural	2,112.80	110	.701	.129	.122		
Metric	2,105.80	120	.701	.124	.123	.000	−.005
Scalar	2,198.10	131	.694	.120	.124	−.007	−.004
Trait–state decomposition							
Four occasions	8,101.90	815	.833	.061	.108		
2018 occasions only	6,332.50	599	.851	.062	.110		
Weighted	7,875.50	815	.833	.061	.107		

Note. CFI = comparative fit index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual. ΔCFI and ΔRMSEA are the index on that row minus the index on the preceding row in the same panel; reference rows are blank. The configural model for the 2018 occasions is the one-factor model with loadings and intercepts free across occasions. All chi-square tests had $p < .001$. $N = 3{,}052$, except the presentation-group models ($N = 2{,}819$; group *n*s = 703, 708, 697, and 711). Mode samples were web ($n = 2{,}360$) and mail ($n = 692$). Robust CFI and RMSEA were not computable for these models; standard indices are reported. No 2018 item intercept was freed. Under the joint ΔCFI, ΔRMSEA, and ΔSRMR criterion (Chen, 2007), no step was rejected, including the scalar step linking 2020 (ΔCFI = −.033), whose RMSEA and SRMR changes stayed within bounds; a CFI-only criterion of .010 would reject that step. Within-occasion correlations of the positive and negative factors were .32, .39, and .39. Trait shares of occasion-factor variance were .82, .83, .86, and .81 for waves fd, fe, and ff (2018) and hc (2020). McDonald's ω and Cronbach's α were .75 and .76 (fd, $n = 2{,}516$), .76 and .78 (fe, $n = 2{,}840$),

.77 and .77 (ff, n = 2,674), and .52 and .63 (hc, n = 2,498). The weighted row uses the design weight rescaled to a mean of 1.

Table 3

Piecewise Latent Growth Model of Life Satisfaction and Regressions on Trait Self-Compassion

Parameter	*b*	*SE*	95% CI	*p*	β
Level at the 2020 knot	70.05	0.87	[68.35, 71.75]	< .001	
Pre-2020 slope	0.40	0.26	[−0.10, 0.90]	.119	
Disruption slope (2020–2021)	−3.11	0.26	[−3.61, −2.61]	< .001	
Recovery slope	0.81	0.17	[0.46, 1.15]	< .001	
Trait self-compassion → level	2.16	0.38	[1.40, 2.91]	< .001	.14
Trait self-compassion → pre-2020 slope	0.25	0.10	[0.05, 0.45]	.015	.12
Trait self-compassion → disruption slope	−0.10	0.31	[−0.71, 0.51]	.746	−.02
Trait self-compassion → recovery slope	−0.29	0.20	[−0.69, 0.11]	.154	−.09

Note. MLR = robust maximum likelihood; POMP = percent of maximum possible; CFI = comparative fit index; TLI = Tucker–Lewis index; RMSEA = root-mean-square error of approximation; SRMR = standardized root-mean-square residual; BIC = Bayesian information criterion. Estimates are from the primary unweighted piecewise model. Life satisfaction is on a 0–100 POMP metric. Slopes are change per year. The disruption slope runs from the start of 2020 to the start of 2022, and the recovery slope is the change after 2022. Indicator intercepts were fixed at 0. N = 3,052. Standard errors are MLR. Fit indices are standard maximum likelihood, $\chi^2(140) = 424.34$, $p < .001$, CFI = .974, TLI = .966, RMSEA = .026, 90% CI [.023, .029], SRMR = .037. Robust CFI, TLI, and RMSEA were numerically unstable and are not interpreted. β is the standardized regression coefficient and is reported for the trait regressions only; the four growth parameters are factor means, for which β does not apply. Bayes factors from BIC comparisons favored the models without the trait paths for the disruption slope, $BF_{01} = 51.52$, and 2020 loneliness, $BF_{01} = 18.42$ (Wagenmakers, 2007).

Table 4

Random-Slope Multilevel Models of Life Satisfaction on Telework Share

Predictor	Model 1	Model 2
Intercept	67.47** (0.51)	67.47** (0.51)
Within-person telework	−0.85 (0.54)	−0.85 (0.54)
Person-mean telework	0.52 (0.35)	0.52 (0.35)
Trait self-compassion	1.58** (0.56)	1.56** (0.56)
Age	0.00 (0.05)	0.00 (0.05)
Education	0.42 (0.48)	0.42 (0.48)
Prepandemic affect	3.99** (0.39)	3.98** (0.39)
Female	2.72** (0.98)	2.72** (0.98)
Within-person telework × trait self-compassion		−0.96 (0.54)

Note. Model 1 is the unmoderated random-slope model. Model 2 adds the cross-level interaction. Entries are unstandardized coefficients with standard errors in parentheses. Within-person telework is the occasion deviation from the person mean. Person-mean telework, age, education, prepandemic affect, and the female indicator were centered, and trait self-compassion was standardized. Life satisfaction and telework were jointly observed on 4,404 person-years from 1,384 persons (1,087 with at least two observations); the 3,484 person-years from 1,080 persons with complete covariates entered the models. The life-satisfaction intraclass correlation was .36. Telework share was observed for 1,410 persons (6,275 reports; 1,164 with at least two reports); the telework-share intraclass correlation was .75. A frequency-weighted sensitivity model was estimated and is not shown. A Bayesian information criterion (BIC) comparison favored Model 1 over Model 2, $BF_{01} = 12.28$ (Wagenmakers, 2007). Simple slopes of within-person telework from Model 2 were $b = 0.10$ ($SE = 0.76$), 95% CI [−1.39, 1.59], $p = .893$ at −1 *SD* of trait self-compassion; $b = -0.85$ ($SE = 0.54$), 95% CI [−1.91, 0.20], $p = .114$ at the mean; and $b = -1.81$ ($SE = 0.77$), 95% CI [−3.31, −0.31], $p = .018$ at +1 *SD*. **$p < .01$.

Table 5

Trait Self-Compassion and the 2024 Technology Outcomes

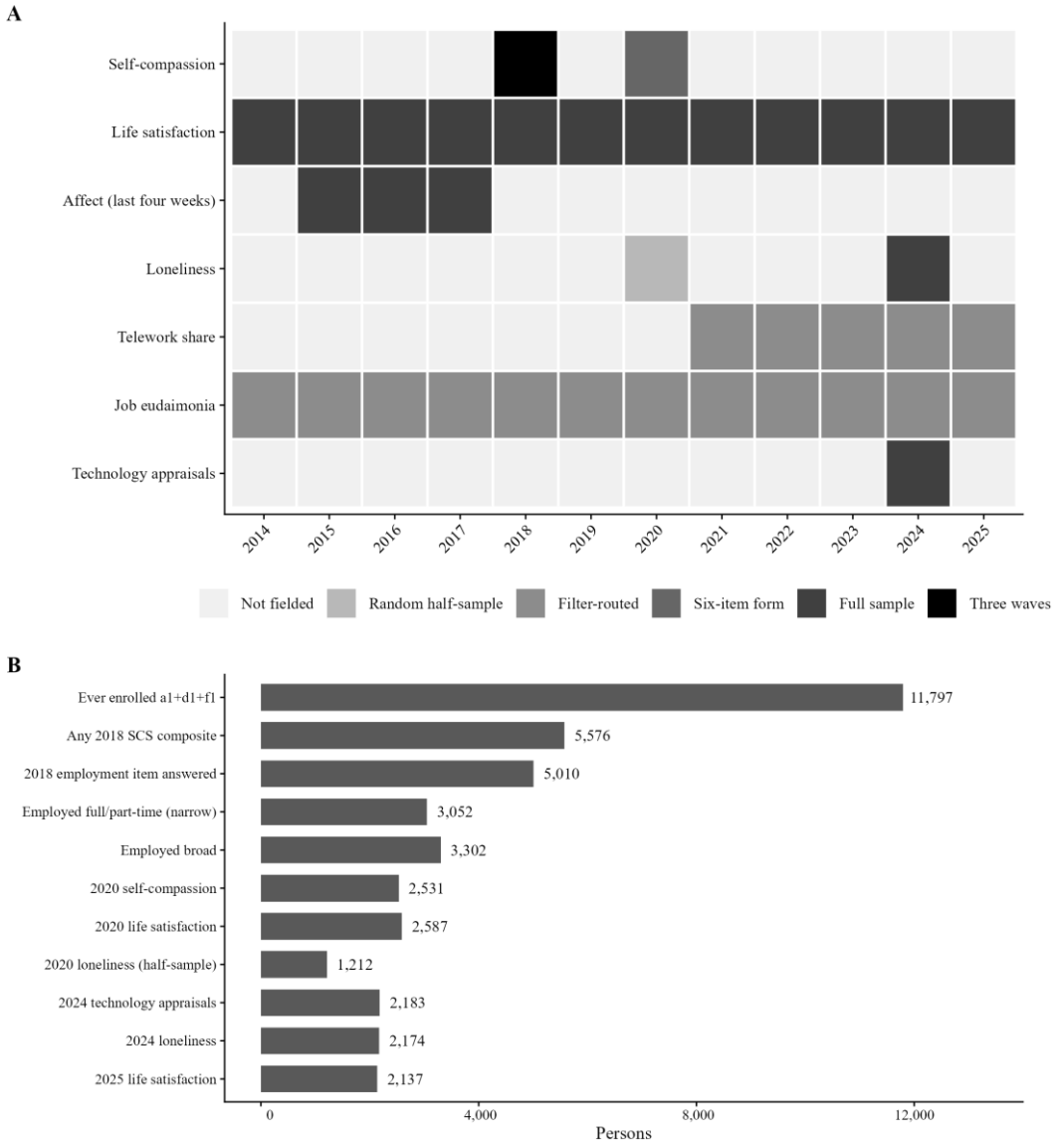
Outcome	*b*	*SE*	95% CI	*p*	β
Digital overwhelm	−0.20	0.02	[−0.24, −0.16]	< .001	−.20
Technology favorability	0.06	0.01	[0.04, 0.09]	< .001	.10
Digitalization job-loss belief	0.02	0.02	[−0.03, 0.07]	.405	.02
Familiarization behavior[a]	0.98	0.05	[0.88, 1.08]	.641	

Note. Estimates are from the primary model. Coefficients are for trait self-compassion, controlling for prepandemic well-being, age, education, and technology affinity. Covariates were centered and trait self-compassion was standardized. β is the standardized coefficient. The three continuous outcomes were estimated jointly with correlated residuals under robust full-information maximum likelihood (N = 3,052); the model is saturated, so no fit indices apply. Bayes factors from Bayesian information criterion (BIC) comparisons (Wagenmakers, 2007): overwhelm $BF_{10} = 1.44 \times 10^{19}$; favorability BF_{10} = 3,179.03; job-loss belief BF_{01} = 38.01; familiarization BF_{01} = 40.73. The job-loss belief is the item hypothesized, on its registry label, as perceived usefulness for job retention (see Measures).

[a] Odds ratio from a complete-case logistic model with heteroskedasticity-consistent standard errors (n = 2,066). On that row the *b* column is the odds ratio, the *SE* is the standard error of the log-odds coefficient (log-odds coefficient = −0.02), and the confidence interval is on the odds-ratio metric. No standardized coefficient was estimated for familiarization.

Figure 1

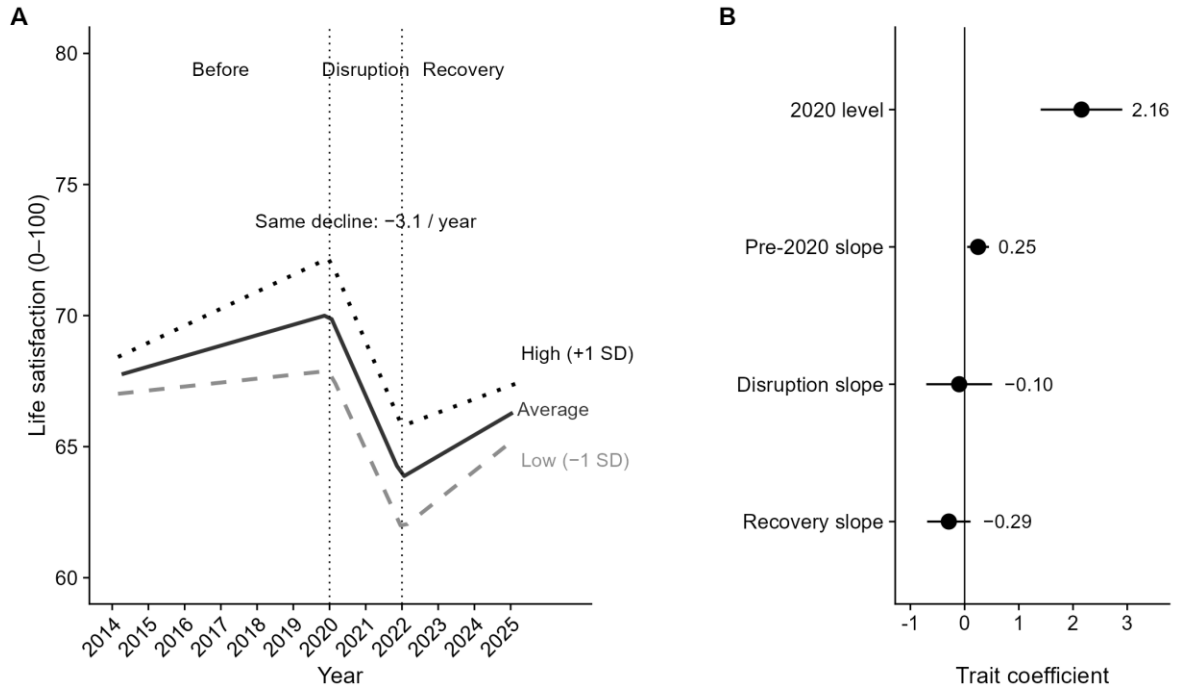
Design Timeline and Sample Flow



Note. Panel A: fielding of self-compassion, life satisfaction, affect, loneliness, telework share, job eudaimonia, and the 2024 technology appraisals from 2014 through 2025, by administration status. Panel B: counts from enrollment to the analytic sample and to later outcomes.

Figure 2

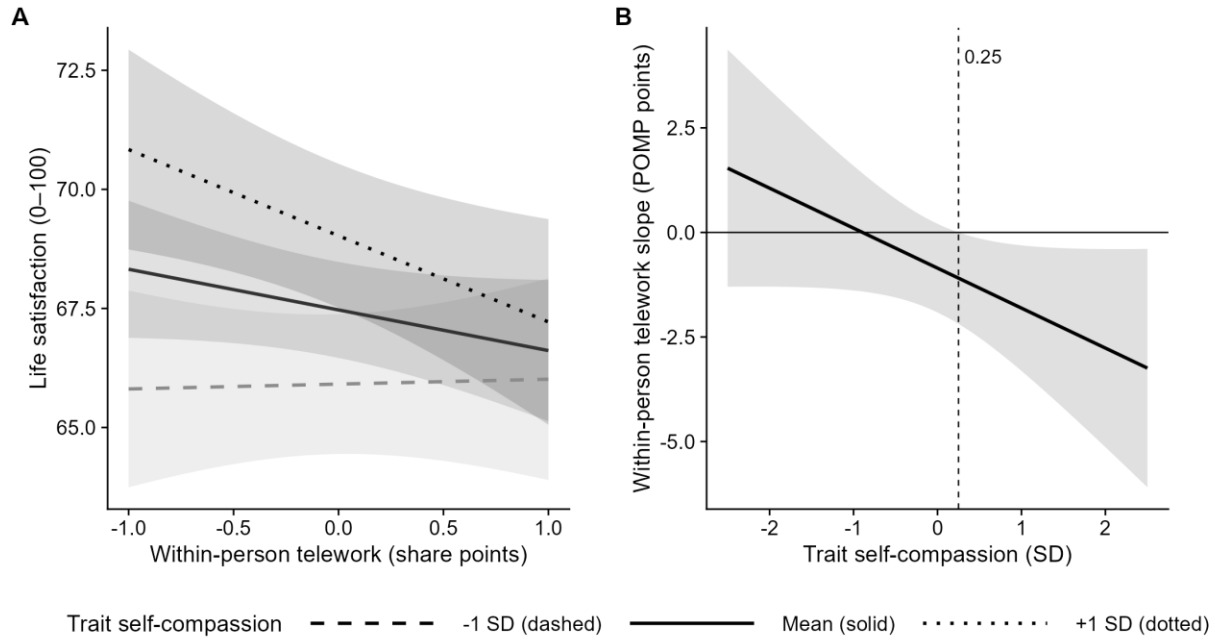
Life Satisfaction Trajectories and Trait Associations With Intercept and Slopes, 2014–2025



Note. Panel A: model-implied life satisfaction at −1 *SD* (dashed), the mean (solid), and +1 *SD* (dotted) of trait self-compassion, other covariates at their means. Life satisfaction is percent of maximum possible (0 = *extremely dissatisfied*, 100 = *extremely satisfied*). The stacked lines are the level association; the shared drop from 2020 to 2022 (−3.1 points per year) is the average decline. Vertical dotted lines mark where the modeled slope is allowed to change. The y-axis is truncated. Panel B: unstandardized coefficients for the standardized trait on the 2020 level (POMP points) and on each slope (POMP points per year). Error bars are 95% confidence intervals; values at the marks are the point estimates. The disruption-slope interval includes zero.

Figure 3

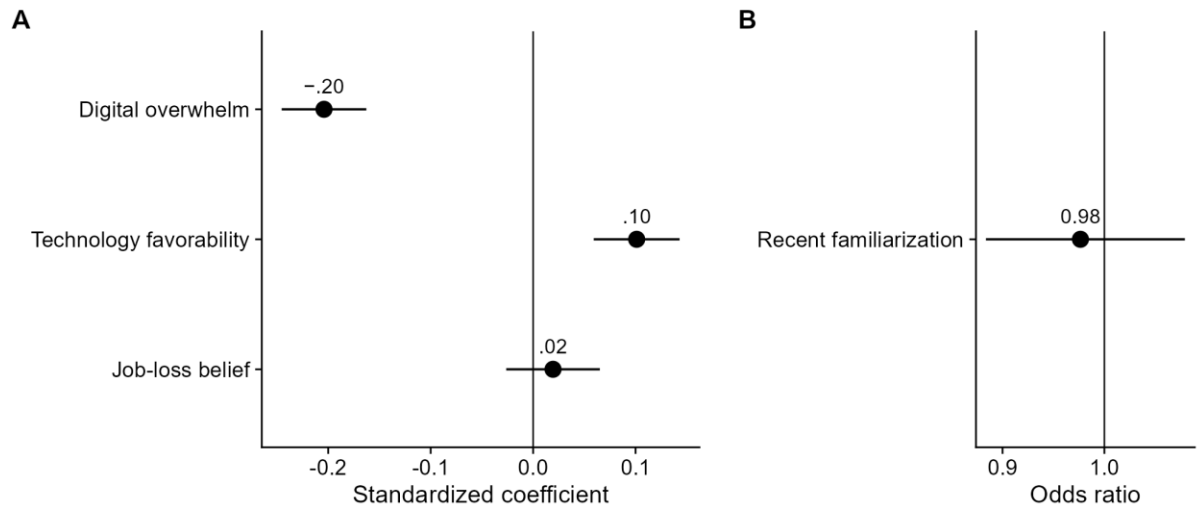
Within-Person Telework and Life Satisfaction by Trait Self-Compassion



Note. Panel A: model-implied life satisfaction across the observed range of person-mean-centered telework share at −1 *SD* (dashed), the mean (solid), and +1 *SD* (dotted) of trait self-compassion. The y-axis is truncated; life satisfaction is scored as percent of maximum possible (POMP; 0–100). Panel B: the within-person telework slope (POMP points per share point) as a function of trait self-compassion, with the Johnson–Neyman bound marked by the dashed vertical line. Bands are 95% Wald intervals from the random-slope model. The interaction was not statistically significant (p = .076); the panels are descriptive.

Figure 4

Trait Self-Compassion Coefficients for the 2024 Technology Outcomes



Note. Panel A: standardized coefficients for trait self-compassion on the three continuous 2024 outcomes, controlling for prepandemic well-being, age, education, and technology affinity. Panel B: the odds ratio for recent familiarization with a new digital technology. Error bars are 95% confidence intervals; values at the marks are the point estimates.

(Supplement: Figure S1 schematic of the structural models; Figure S2 within/between telework decomposition; attrition analyses, sensitivity models, full invariance tables, rationale for estimators considered and not adopted, full caring-sector and component-specific estimates, elastic net tuning grid with per-outcome out-of-fold and single-split metrics, item wordings in English and German.)